Deliverable 2: Interim Progress Report

Cybersecurity Analytics Class Project

Deliverable	Deliverable 2: Interim Progress Report 1	**Selected Track**	Fraud
Project Title	Deepfake Recognition	**Team Name**	Team 6
Team Members	Flemming Grabowski, Payel Sarkar	**Submission Date**	October 11, 2026
Repository	https://github.com/payelsarkarr/IS665-Team6-Deepfake-Recognition	**Dataset Link**	FaceForensics++ (official access granted): https://github.com/ondyari/FaceForensics . Interim analysis on the Kaggle re-upload xdxd003/ff-c23 (see Section 1).

Executive Status Snapshot

Item	Current status
Overall status	At risk. The data pipeline, quality checks, EDA and two baselines are done on a 1,250-video subset, but we started late: the proposal milestones for the data audit (Sep 27) and the baseline CNN (Oct 4) slipped, and the CNN baseline has not started. The official FF++ servers were unreachable on Oct 7-11.
Most important accomplishment	A reproducible Colab pipeline (manifest, integrity and leakage checks, EDA, baselines) run on 1,250 videos. Key finding: simple frame-difference features barely beat chance (ROC-AUC 0.53), so temporal evidence has to be learned.
Most important risk	Schedule. Preprocessing is CPU-bound (about 3 s per video in Colab), so the full 5,000 videos need about 4 hours before any CNN training. Mitigation: cache face crops once and move face detection to the GPU.
Next milestone	Face crops for all 5,000 videos and the simple CNN baseline (B2) by Oct 18, 2026 (Section 7).

1. Project Status and Changes from Proposal

Completed. We received access to the official FaceForensics++ (FF++) release. On Oct 7-11, 2026 all three official download servers refused connections, so this report uses a public Kaggle re-upload of the same c23 release, checked against the official file lists (Section 3). Our Colab pipeline (FFpp_EDA_Colab.ipynb, ff_eda.py) builds a video manifest, assigns the official splits, checks integrity, duplicates and leakage, extracts face clips with temporal features and runs two baselines. The CNN baseline has not started.

Changes in response to feedback.

- **Dataset source.** After the Professor's comment on the Kaggle copy in the proposal, we obtained official access. Because of the outage, this analysis still uses a Kaggle copy and will be re-run on the official download.
- **More data.** The project will use all 5,000 videos instead of 400. This report analyses a 1,250-video subset because preprocessing on Colab CPUs is slow (Section 7).
- **Temporal architecture.** Fakes are generated frame by frame, so transitions may show flicker or unstable blending. MobileNetV2 stays as per-frame encoder; a GRU head reads clips of consecutive face crops, compared against the frame-only model in an ablation.
- **Concrete explainability.** Two outputs only: a per-clip score timeline and Grad-CAM heatmaps for the highest-scoring frames.
- **Tooling.** Ultralytics HUB was dropped (shut down July 31, 2026; built for YOLO).

Feasibility. The scope is feasible but the schedule is tight (Section 7). Success criteria are unchanged (fake recall ≥ 0.80, false-positive rate ≤ 0.20).

2. Dataset Description and Data Dictionary

Source and context. FF++ (Rössler et al., 2019) contains 1,000 original sequences from 977 YouTube videos (mostly TV news), each manipulated by two identity swaps (Deepfakes, FaceSwap) and two expression reenactments (Face2Face, NeuralTextures). Fake 000_003 uses video 000 as target and 003 as source. We use the h264 c23 version; there is no audio and no time dimension. Videos are not stored in the repository (Terms of Use).

Records, labels and classes. The full release has 5,000 videos, 1,000 real and 4,000 fake (1:4); the Kaggle copy has all of them plus two unused folders (FaceShifter, DeepFakeDetection). **This report analyses 1,250 videos:** originals 000-249 and, per method, the 250 fakes whose target is one of them (still 1:4). Labels and splits are per video; the model input is a clip of 16 consecutive face crops (2 per video). The target is binary (0 = real, 1 = manipulated); the method is kept for error analysis. With the official splits and the leakage exclusion (Section 3) the subset has 970 training (194 real), 101 validation (21 real) and 171 test videos (35 real); see Figure 5 in Appendix B. The data dictionary (12 fields) is in Appendix A.

Video properties. 15-60 fps (624 videos at 25 fps, 560 at 30 fps); resolutions 576×480 to 1920×1080 (443 at 640×480, 350 at 1280×720). Median length 455 frames (real) vs. 407 (fake).

Limitations. Mostly frontal news-studio faces; methods from 2017-2019 (no diffusion or real-time swaps); no demographic metadata; unofficial copy for now; only 21 real validation videos in the subset.

3. Data-Quality Assessment

All checks run at the video level, because the video is the unit that must not cross splits. Values come from quality_report.json of the 1,250-video run unless stated otherwise.

Table 2. Data-quality evidence and treatment (1,250-video subset).

Issue checked	Evidence / magnitude	Fix/Mitigation	Remaining risk
Missing / unreadable; source integrity	0 unreadable, 0 too short, 0 without split; all 1,250 file names in the official list	None needed; checks stay in the pipeline	Unofficial copy; re-run on official data
Duplicates	0 exact (SHA-256); 0 near-duplicates across targets (frame hash)	Real/fake of one target kept in one split, not removed	Hash uses one frame
Inconsistent values / outliers	7 frame rates (15-60 fps); 576×480 to 1920×1080; 65 length outliers (IQR)	Crops resized to 224×224; max 2 clips per video	16 frames = 0.27-1.07 s; fixed-fps resampling planned
Face detection	Face in 99.4% of clips (99.0-99.6% per method); >1 face in 25.6% of videos	Clips without face dropped; largest face kept	False or second-person detections
Class imbalance	1:4 (250 vs. 1,000)	Class weights (train only); recall, FPR, PR-AUC	Real-world prevalence is far lower
Label quality	By construction. Spot check of 5 NeuralTextures pairs: change barely visible; Deepfakes blur, FaceSwap edges visible (Fig. 4)	Kept; results per method	Subtle methods lower recall
Leakage across splits	22 YouTube videos gave 45 sequences; 7 groups cross official splits. Visual check: different people in all 7 pairs, but same studio (once two sisters)	Conservative: 7 val/test IDs and their fakes excluded (8 videos here, 63 in full set); split by video ID	No identity leak; shared studio/logo could be a shortcut
Shortcut features	Real videos longer (455 vs. 407 frames) and larger (1.33 vs. 1.19 MB) (Fig. 2)	Length, size, bitrate, resolution never used as inputs	Re-encoding traces may differ

4. Exploratory Data Analysis

The EDA asks two questions: do trivial shortcuts separate the classes, and do frame-to-frame changes differ between real and fake faces? The split distribution (Figure 5) and example face crops (Figure 4) are in Appendix B. Because real videos are only 20% of every split, calling everything fake already gives 79% accuracy (Section 6), so we report recall and FPR.

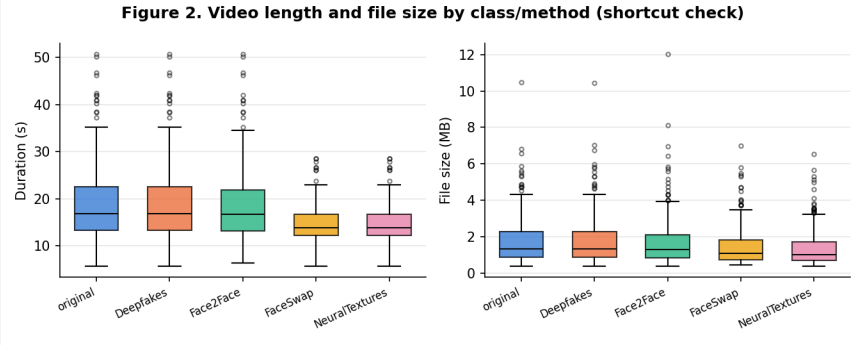


Figure 2. Video duration and file size by method (image title: shortcut check). Takeaway: FaceSwap and NeuralTextures videos are shorter and smaller than originals, so container properties would leak the label.

Deepfakes and Face2Face match their originals in length, but FaceSwap and NeuralTextures are clearly shorter (median about 14 s vs. 17 s) and smaller. A model using length, size or bitrate would learn the generation pipeline, and an attacker could evade it by trimming or re-encoding. These properties are therefore excluded from all model inputs.

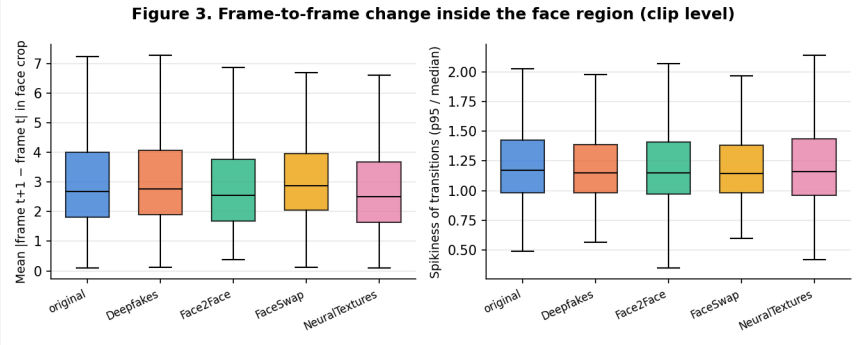


Figure 3. Mean frame-to-frame change inside the face crop (left) and spikiness of these changes within a 16-frame clip (right), per method. Takeaway: the distributions of all five classes almost completely overlap.

Median change is 2.5-2.9 grey levels and spikiness about 1.15 for every class. Global pixel differences are dominated by natural motion (head, speech, camera), not by manipulation artifacts. Transitions may still carry evidence, but not in one global statistic, as the baseline confirms (Section 6). Temporal cues must therefore be learned from aligned face crops (CNN features fed to the GRU) rather than hand-crafted.

5. Data Preparation and Feature Engineering

The pipeline runs per video. The only fitted transform (the B1 scaler) is fitted on the training split inside a scikit-learn pipeline (StandardScaler); CNN inputs will be scaled to [-1, 1] by a fixed formula; seed 665.

FFpp_EDA_Colab.ipynb reproduces all results from the raw videos. Augmentation (flip, brightness, JPEG re-compression, one setting per clip, training only) is planned for the CNN.

Table 3. Preparation steps and engineered features.

Step / feature	Implementation	Rationale	Leakage or bias control
Split assignment	Official FF++ split files; fakes take the split of target and source ID; leakage exclusion (Section 3)	Random frame or video splits would place the same person and background in train and test	Split by ID before any frame is extracted
Clip sampling	2 evenly spaced clips of 16 consecutive frames per video	Consecutive frames are needed to see transitions	Clips of one video never cross splits
Face detection and crop	OpenCV Haar detector on frames downscaled to 640 px (same hit rate as full resolution on 30 frames, 2× faster); largest face; box ×1.3, fixed per clip	Focus on the manipulated region; a fixed box avoids adding artificial jitter	Same procedure for real and fake
Temporal features (created)	diff_mean, diff_std, diff_max, diff_p95_ratio from consecutive crop differences; sharpness (variance of Laplacian)	Simple, interpretable measure of transitions (feedback)	Computed per clip; no labels used
Removed features	Length, file size, bitrate, resolution, fps, method name	Would let the model learn encoding or label shortcuts (Figure 2)	Kept in the manifest for quality control only

6. Baseline Model or Analytical Benchmark

Evaluated per video on the subset's validation split (21 real, 80 fake); the test split was not used. **B0** predicts "fake" for every video. **B1** is a logistic regression on the seven clip features of Table 3 (class_weight = balanced; clip probabilities averaged per video, threshold 0.5).

Table 4. Baseline results on the validation split of the subset (video level, n = 101).

Metric	Baseline result	Interpretation
Precision (fake)	B0: 0.792 B1: 0.808	B1 barely above the share of fakes
Recall (fake)	B0: 1.000 B1: 0.738	Below the 0.80 target; per method: Deepfakes 0.85, NeuralTextures 0.80, Face2Face 0.70, FaceSwap 0.60
False-positive rate	B0: 1.000 B1: 0.667	14 of 21 real videos flagged; far above the 0.20 target
F1 (fake)	B0: 0.884 B1: 0.771	B0 looks best only because of the imbalance
ROC-AUC / PR-AUC	B0: 0.500 / 0.792 B1: 0.528 / 0.811	B1 ranks videos almost at chance level

What this establishes. Hand-crafted global temporal features carry almost no signal (ROC-AUC 0.53), consistent with Figure 3; F1 alone would make the trivial B0 look strong. This is the floor for the simple CNN (B2) and MobileNetV2. With 21 real videos the estimates are uncertain (one video ≈ 5 points of FPR); they will be repeated on the full data.

7. Challenges and Revised Plan

Challenges. (a) Official servers down; results will be re-run on official data. (b) Preprocessing is CPU-bound: about 3 s per video in Colab (216 ms per face detection), roughly 4 hours for 5,000 videos; we will cache crops once and test a GPU face detector. (c) We are behind the proposal schedule. (d) Natural motion masks temporal artifacts (Figure 3).

Open decisions. Framework (Keras or PyTorch), fixed-fps resampling, clip length (8/16/32 frames), GRU vs. temporal convolution.

Revised plan. By Oct 18: face crops for all 5,000 videos cached; simple CNN baseline (B2) on validation. By Oct 25: MobileNetV2 frame-only vs. MobileNetV2 + GRU (temporal ablation). By Nov 1: leave-one-method-out test, score timeline and Grad-CAM. Nov 4: Interim Report 2 (as in the proposal). If the official servers are back, the data step is repeated on the official download first.

References

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Team Contribution Statement

Team member	Responsibilities and completed work	Evidence / artifact
Flemming Grabowski	Ran the Colab pipeline on the data (loading, quality checks, EDA, baselines)	Notebook FFpp_EDA_Colab.ipynb and outputs (see repository)
Payel Sarkar	Performed the visual leakage and label checks, revised the report	Report

This milestone was harder than expected because it overlapped with our midterm exams, so we completed less than we had planned and started the data work later than scheduled. Our coordination itself worked well: we met before class each week, often while studying together, to review what had been done and agree on the next steps. These short weekly check-ins kept us aligned, and we will continue them. To catch up, we will also use them to set concrete tasks and deadlines for the milestones in Section 7.

AI-Use Disclosure

Required: Disclose any generative-AI tools used for brainstorming, coding, editing, analysis, or slide creation. You remain responsible for the accuracy, originality, security, and reproducibility of all submitted work.

Tool	How it was used	What the team verified or changed
Claude (Anthropic)	Drafted the report text and structure, along with the Colab notebook, using a detailed prompt that outlined our requirements and acceptance criteria. (iterative process)	We ran the notebook on the data, reviewed all the figures, and completed the visual checks. We corrected the code, examined the data and its structure, and finalized the report. We also corrected an assumption in the draft: the seven overlapping pairs represent different people.

Submission Checklist

Check	Requirement
[x]	Report reflects work completed by the deadline.
[x]	All tables and figures have numbers, captions, and interpretations.
[x]	Data-quality findings include quantitative evidence.
[x]	Baseline results use a clearly identified validation set.
[x]	Repository contains the code or notebooks referenced.
[x]	Contribution statement and AI-use disclosure are complete.

Rubric Self-Check

Criterion	Points	Where the evidence appears
Status, scope, and project management	1	Snapshot, Section 1
Dataset description and data dictionary	2	Section 2, Appendix A (Table 1)
Data-quality assessment	2	Section 3, Table 2
EDA and interpretation	2	Section 4, Figures 2-3, Appendix B (Figures 4-5)
Preparation, features, and baseline	2	Sections 5-6, Tables 3-4
Challenges, plan, and report quality	1	Section 7, contribution statement, repository link

Appendix A. Data Dictionary

Table 1. Data dictionary.

Variable / field	Type	Meaning	Role in analysis
video_id	String (ID)	Unique file key, e.g. 000 or Deepfakes/000_003	ID, not a feature
label	Binary	0 = original, 1 = manipulated	Target
method	Categorical (5 levels)	original, Deepfakes, Face2Face, FaceSwap, NeuralTextures	Error analysis; excluded from input
target_id, source_id	String (3-digit ID)	Sequence providing frames / identity (source empty for real)	Grouping key for split and leakage checks
youtube_id	String	YouTube video the sequence was cut from (conversion_dict.json)	Leakage check only
split	Categorical	train / val / test from official split files	Partition
n_frames, fps, width, height, duration_s, file_mb	Numeric	Container metadata read with OpenCV	Quality control; excluded from input
sha256, dhash_mid	String	File hash; perceptual hash of a middle frame	Duplicate detection
face_found, face_rate	Binary / ratio 0-1	Face detected in clip; share of sampled frames with a face	Quality filter; B1 feature
diff_mean, diff_std, diff_max, diff_p95_ratio	Numeric (grey levels)	Statistics of mean absolute change between consecutive face crops in a clip	Engineered temporal features (B1)
sharp_mean, sharp_std	Numeric	Variance of Laplacian per crop and its variation across the clip	Engineered features (B1)
face crop clip	uint8 tensor 16×224×224×3	Fixed-box face crops of consecutive frames	Input to CNN + GRU (planned)

Appendix B. Example Face Crops

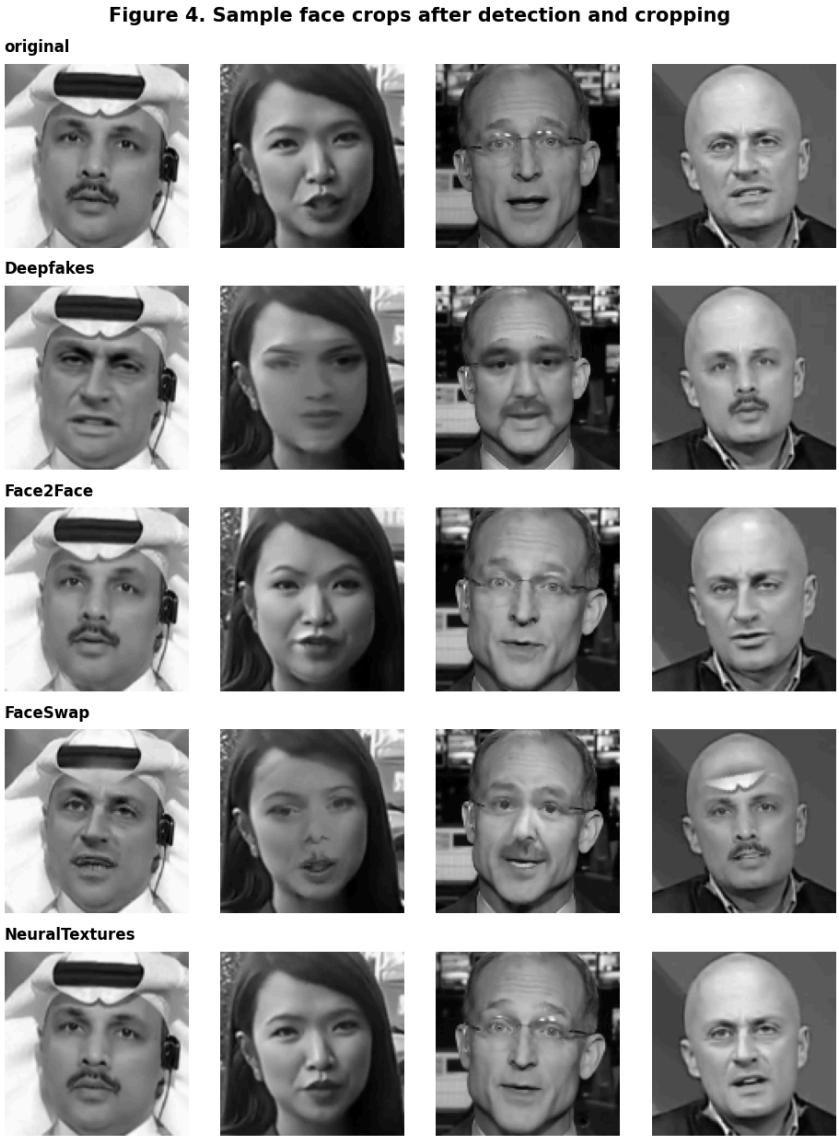


Figure 4. Face crops of the same four target videos for each class. Deepfakes crops are visibly blurred, FaceSwap shows blending edges (fourth column, forehead), while Face2Face and NeuralTextures changes are hard to see in a single frame.

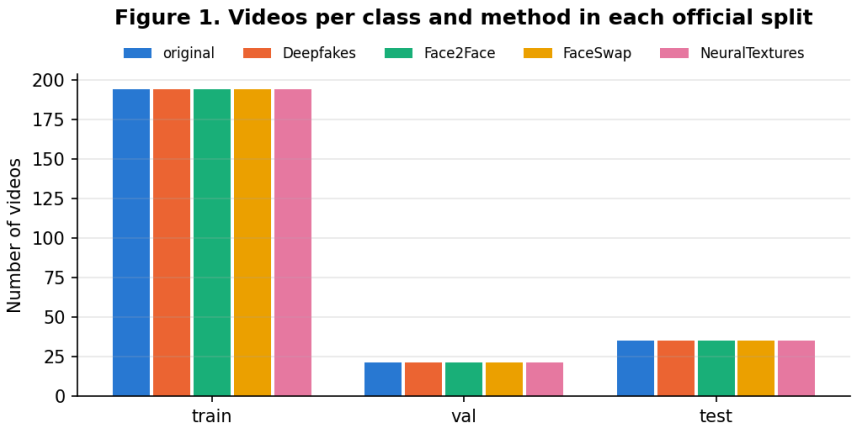


Figure 5 (image title: Figure 1). Videos per class and method in the official splits of the 1,250-video subset after leakage exclusion. Every split keeps the 1:4 ratio; validation has only 21 real videos.